

**Important and Solved Questions for Mid Term Exam**

L02_Q01: Find value of following trigonometric function at $x = \frac{\pi}{2}$ using subs command
Cos(x), Sin(x), Tan(x). **(FOLLOW P#37-38 - LECT#02)**

```
clear all
close all
clc
syms x a
a=cos(x)
subs(a,x,sym('pi/2'))
a=sin(x)
subs(a,x,sym('pi/2'))
a=tan(x)
subs(a,x,sym('pi/2'))
```

L02_Q02: Find value of following trigonometric function at $x = \frac{\pi}{2}$ using sym command
Cos(x), Sin(x), Tan(x). **(FOLLOW P#37-38 - LECT#02)**

```
close all
clear all
clc
cos(sym(pi/2))
sin(sym(pi/2))
tan(sym(pi/2))
```

L02_Q03: Find the value of the following using subs command. $u(x,y) = 2x + xy^2 + 3y$
at $x=5$, $f(x) = \sin x$ at $x = \frac{\pi}{2}$

(Follow P#036-42 - LECT#02)

(ABU SULTAN MID TERM Q#3(a) – 26 DEC 2019 – 12:00 pm)

```
clear all
close all
clc
syms x y u f
u=2*x+x*y^2+3*y;
subs(u,x,5)
f=sin(x)
subs(f,x,pi/2)
```

L02_Q04: Find the value of the following polynomials at $x=2$ using subs command.

$p(x,y) = x^2 + 3y^2x + 10$, $u(x) = x^3 + 4x + 10$ **(Follow P#036-42 - LECT#02)**

```
clear all
close all
clc
syms x y p u
p=x^2+3*y^2*x+10
subs(p,x,2)
u=x^3+4*x+10
subs(u,x,2)
```

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L03_Q05: Find the solution of $x^2 - y = 2$ and $y - 2x = 5$ (P#066-68 - LECT#03)

close all

clear all

clc

syms x y

[x,y]=solve('x^2-y=2','y-2*x=5')

x(1)

y(1)

L03_Q06: Find solution of equation $2x^3 + x^2 + 3x + 4 = 0$ (FOLLOW P#066-68 - LECT#03) (ABU SULTAN MID TERM Q#3(b) – 26 DEC 2019 – 12:00 pm)

close all

clear all

clc

syms x

solve('2*x^3+x^2+3*x+4=0')

L04_Q07: Solve by solve command. $2x + 7y = 3$, $3x^2 + 2y = 1$. Also plot graph for $3x^2 + 2y = 1$. Name it parabola, give x-label and y-label also.

(FOLLOW P#066-71 - LECT#03) and (FOLLOW P#82,86 - LECT#04)

clc

close all

clear all

syms x y

solve('2*x+7*y=3','3*x^2+2*y=1')

ezplot('3*x^2+2*y=1',[-2,2])

xlabel('x-axis')

ylabel('y-axis')

title('Parabola')

L04_Q08: Plot the graph of e^{-x} and $\sin(x)$ on the interval $[0, 8]$ using ezplot, hold on and hold off commands. Label the horizontal axis with x and vertical axis with y. Finally gives the title "Multiple Graphs". (P#90-92 - LECT#04)

%Plot the graph of e^{-x} and $\sin(x)$

%on the interval $[0, 8]$ using ezplot, hold on and hold off commands.

%Label the horizontal axis with x and vertical axis with y.

%Finally gives the title "Multiple Graphs"

clc

clear all

close all

syms x

ezplot(exp(-x),[0,8])

hold on

ezplot(sin(x),[0,8])

hold off

xlabel('x')

ylabel('y')

title('Multiple Graphs')

grid

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L04_Q09: Plot the graph of e^{-x} and $\sin(x)$ on the interval $[0, 8]$ using plot, hold on and hold off commands. Label the horizontal axis with x and vertical axis with y. Finally gives the title "Multiple Graphs". (P#90-92 - LECT#04)

%Plot the graph of e^{-x} and $\sin(x)$ on the interval $[0, 8]$ using plot, hold on and hold off.

%Label the horizontal axis with x and vertical axis with y.

%Finally gives the title "Multiple Graphs"

clc

clear all

close all

X=0:0.1:8;

plot(X,exp(-X))

hold on

plot(X,sin(X))

hold off

xlabel('x')

ylabel('y')

title('Multiple Graphs')

grid

L04_Q10: Plot the parametric graph of the circle with center at (0; 0), radius 1, and with increment of 0:01 in the domain set. Label the horizontal axis with x and vertical axis with y. Make the shape of the graph square. Keeps the title of the graph "Parametric plot of the circle". (P#98-100 - LECT#04)

%Plot the parametric graph of the circle with center at (0; 0),

% with radius 1 and with increment of 0:01 in the domain set.

clc

clear all

close all

t=0:0.01:1;

plot(cos(2*pi*t),sin(2*pi*t))

xlabel('x')

ylabel('y')

title('Parametric Plot of the Circle')

grid

L04_Q11: Plot the contour plots of the expression $x^2 + y^2$ using meshgrid and contour commands. Label the horizontal axis with x and vertical axis with y. Make the shape of the graph square. Keeps the title of the graph "Contour plotting of squares".

(P#102-104 - LECT#04)

%Plot the contour plots of the expression $x^2 + y^2$ using meshgrid and contour commands.

%Label the horizontal axis with x and vertical axis with y.

%Make the shape of the graph square.

%Keeps the title of the graph "Contour plotting of squares".

clc

clear all

close all

[X,Y]=meshgrid(-2:0.1:2,-2:0.1:2);

contour(X,Y,X.^2+Y.^2)

axis square

xlabel('x')

ylabel('y')

title('Contour Plotting of Squares')

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L04_Q12: Plot the graph of $x^3 + x^2 + 4$ on the interval $-2, 2]$ using `ezplot`, hold on and hold off commands. Label the horizontal axis with x and vertical axis with y . Finally gives the title " $x^3 + x^2 + 4$ ". (Follow P#90-92 - LECT#04)

%Plot the graph of x^3+x^2+4 on the interval $-2, 2]$ using `ezplot`, hold on and hold off commands.

%Label the horizontal axis with x and vertical axis with y .

%Finally gives the title " x^3+x^2+4 ".

`clc`

`clear all`

`close all`

`syms x`

`x=x^3+x^2+4`

`ezplot(x,[-2,2])`

`axis square`

`xlabel('x')`

`ylabel('y')`

`title('x^3+x^2+4')`

L04_Q13: Plot the graph of the function $\sin(x) + \cos(x)$ on the interval $[0; 1]$, using `plot` command. Label the horizontal axis with x and vertical axis with y . Make the shape of the graph square. Keeps the title of the graph "The Graph of $\sin(x) + \cos(x)$ ". Keeps the layout of the graph as grid. (Follow P#98-100 - LECT#04)

%Plot the graph of the function $\sin(x) + \cos(x)$ on the interval $[0; 1]$, using `plot` command.

%Label the horizontal axis with x and vertical axis with y .

%Make the shape of the graph square.

%Keeps the title of the graph "The Graph of $\sin(x) + \cos(x)$ ".

%Keeps the layout of the graph as grid.

`clc`

`clear all`

`close all`

`X=0:0.01:1;`

`plot(X,(sin(X)+cos(X)))`

`xlabel('x')`

`ylabel('y')`

`axis square;`

`title('The Graph of  $\sin(x) + \cos(x)$ ');`

`grid`

L04_Q14: Plot the multiple curves of the functions $\sin(x) + \cos(x)$ and $e^{-x} + x^2$ on the interval $[0; 4]$, using `ezplot` command. Label the horizontal axis with x and vertical axis with y . Make the shape of the graph square. Keeps the title of the graph "The Graph of Multiple curves using `ezplot` command ". Keeps the layout of the graph as grid.

(Follow P#90-92 - LECT#04)

%Plot the multiple curves of the functions $\sin(x) + \cos(x)$ and $e^{-x} + x^2$

%on the interval $[0; 4]$, using `ezplot` command.

%Label the horizontal axis with x and vertical axis with y .

%Make the shape of the graph square.

%Keeps the title of the graph "The Graph of Multiple curves using `ezplot` command ".

%Keeps the layout of the graph as grid.

`clc`

`clear all`

`close all`



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```
syms x
ezplot('sin(x)+cos(x)',[0,4])
hold on
ezplot('exp(-x)+x^2',[0,4])
xlabel('x')
ylabel('y')
axis square
title('The Graph of Multiple curves using ezplot command ')
grid
```

L04_Q15: Plot the contour plots of the circles x^2+y^2 of radius $1, 2, \sqrt{2}, \sqrt{3}$. Label the horizontal axis with x and vertical axis with y. Make the shapes of the graph square. Keeps the title of the graph “Contour Plots of Circles”. Keeps the layout of the graph grid.

(P#109 - EXERCISE – LECT#04)

```
%Plot the contour plots of the circles x2+y2 of radius 1,2,?2,?3
%Label the horizontal axis with x and vertical axis with y.
%Make the shapes of the graph square.
%Keeps the title of the graph “Contour Plots of Circles”.
%Keeps the layout of the graph grid.
clc
clear all
close all
syms a b
[X,Y]=meshgrid(-2:0.1:2,-2:0.1:2);
contour(X,Y,X.^2+Y.^2-1);
contour(X,Y,X.^2+Y.^2-2);
contour(X,Y,X.^2+Y.^2-sqrt(2));
contour(X,Y,X.^2+Y.^2-sqrt(3));
axis square
xlabel('x-axis')
ylabel('y-axis')
title('Contour Plots of Circles')
grid
```

L04_Q16: Plot the graph of $\sin(x)+\cos(x)$ and $e(-x)+x^2$ on the interval $[0, 4]$ using plot, hold on and without use of hold on and hold off commands. Label the horizontal axis with x and vertical axis with y. Finally gives the title “The Graph of Multiple Curves using plot command”. **(P#108 - EXERCISE – LECT#04)**

```
%Plot the graph of sin(x)+cos(x) and exp(-x)+x^2
%on the interval [0, 4] using plot command and without use of hold on and hold off
commands.
%Label the horizontal axis with x and vertical axis with y.
%Finally gives the title “The Graph of Multiple Curves using plot command”.
%Keeps the layout of the graph as grid.
clc
clear all
close all
x=0:0.1:4;
f1=sin(x)+cos(x);
f2=exp(-x)+x.^2
plot(x,f1,x,f2)
xlabel('x')
```

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```
ylabel('y')  
title('The Graph of Multiple Curves using plot command')  
grid
```

L05_Q17: Find the first-order, second-order, and fourth-order derivatives of the function $f(x) = x^{10} + 2 * (x^2) + \sin(x)$ Using diff command. Plot the multiple graphs of the outputs using ezplot command on the interval $[-\pi, \pi]$. Label the horizontal axis with x and vertical axis with y. Keeps the title "Multiple plot of the function and its derivatives". All the coding should be written using Script M-File.

(P#144 - EXERCISE – LECT#05)+ (P#174 - EXERCISE – LECT#06)

```
%Find the first-order, second-order, and fourth-order derivatives of the  
function f(x)=x^10+2*(x^2)+Sin(x) Using diff command.  
%Plot the multiple graphs of the outputs using ezplot command on the interval [-?,?].  
%Label the horizontal axis with x and vertical axis with y.  
%Keeps the title "Multiple plot of the function and its derivatives".  
%All the coding should be written using Script M-File  
clc  
clear all  
close all  
syms x  
f=x^10+2*x^2+sin(x);  
f1=diff(f,1)  
f2=diff(f,2)  
f4=diff(f,4)  
ezplot(f,[-pi,pi])  
hold on  
ezplot(f1,[-pi,pi])  
hold on  
ezplot(f4,[-pi,pi])  
hold off  
xlabel('x-axis')  
ylabel('y-axis')  
title('Multiple plot of the function and its derivatives')  
grid
```

L05_Q18: Find the first-order, fourth-order, and fifth-order derivatives of the function $f(x) = \ln x + \sin(x)$ using diff command. Plot the multiple graphs of the outputs using plot command on the interval $[0, 1]$ and without using hold on and hold off commands Label the horizontal axis with x and vertical axis with y. Keeps the title "Multiple plot of the function and its derivatives". All the coding should be written using Script M-File.

(P#145 - EXERCISE – LECT#05) + (P#175 - EXERCISE – LECT#06)

```
%Find the first-order, fourth-order, and fifth-order derivatives of the function  
%f(x)=ln x+Sin(x)using diff command.  
%Plot the multiple graphs of the outputs using plot command on the interval [0, 1]  
%and without using hold on and hold off commands.  
%Label the horizontal axis with x and vertical axis with y.  
%Keeps the title "Multiple plot of the function and its derivatives".  
%All the coding should be written using Script M-File  
clc  
close all  
clear all  
syms x
```



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```
f=log(x)+sin(x)
f1=diff(f,1)
f4=diff(f,4)
f5=diff(f,5)
x=0:0.1:1;
f11=eval(vectorize(f1));
f44=eval(vectorize(f4));
f55=eval(vectorize(f5));
plot(x,f11,x,f44,x,f55)
xlabel('x-axis')
ylabel('y-axis')
title('Multiple plot of the function and its derivatives')
grid
```

L05_Q19: Find the first-order, fourth-order, and fifth-order derivatives of the function $f(x) = \ln x * y + \sin(x^2)$ using diff command. Plot the multiple graphs of the outputs using plot command on the interval [0, 1] and without using hold on and hold off commands Label the horizontal axis with x and vertical axis with y. Keeps the title "Multiple plot of the function and its derivatives". All the coding should be written using Script M-File.

(P#145 - EXERCISE – LECT#05) + (P#176 - EXERCISE – LECT#06)

```
%Find the first-order, fourth-order, and fifth-order derivatives of the function
%f(x)=ln x*y+Sin(x^2) using diff command.
%Plot the multiple graphs of the outputs using plot command on the interval [0, 1]
%and without using hold on and hold off commands.
%Label the horizontal axis with x and vertical axis with y.
%Keeps the title "Multiple plot of the function and its derivatives".
%All the coding should be written using Script M-File
clc
clear all
close all
syms x y
f=log(x*y)+sin(x^2)
d1=diff(f,1)
d4=diff(f,4)
d5=diff(f,5)
x=0:0.1:1;
d11=eval(vectorize(d1));
d44=eval(vectorize(d4));
d55=eval(vectorize(d5));
plot(x,d11,x,d44,x,d55)
xlabel('x-axis')
ylabel('y-axis')
grid
title('Multiple plot of the function and its derivatives')
```

L05_Q20: Find the derivative of the following function with respect to x, and t variables using diff command. Moreover plot the multiple graph of the function and their derivatives by defining X and Y using meshgrid command, Then define Z and create a surface plot: $f(x, t) = xe^{-x^2-t^2}$ $x, t \in [-2, 2]$.

(P#122 – LECT#05)

```
%Find the derivative of the following function with respect to x, and t variables using diff
command.
```

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%Moreover plot the multiple graph of the function and their derivatives by defining X and Y using meshgrid command.

%Then define Z and create a surface plot:

%function is $f(x,t)=xe^{(-x^2-t^2)}$ $x,t \in [-2,2]$

clc

clear all

close all

syms x t

$f=x*\exp(-x^2-t^2);$

$f1=\text{diff}(f,x)$

$f2=\text{diff}(f,t)$

$[x,t]=\text{meshgrid}(-2:0.2:2,-2:0.2:2);$

$ff=\text{eval}(\text{vectorize}(f));$

$f11=\text{eval}(\text{vectorize}(f1));$

$f22=\text{eval}(\text{vectorize}(f2));$

$\text{surf}(x,t,ff)$

hold on

$\text{surf}(x,t,f11)$

hold on

$\text{surf}(x,t,f22)$

hold off

L05_Q21: Find the first-order, second-order, and fourth-order derivatives of the function $f(x) = 2xy + (xy)^3$ with respect to variable y using diff command.

%Find the first-order, second-order, and fourth-order derivatives of the function

% $f(x)=2xy+(xy)^3$ with respect to variable y using diff command.

syms x y

$f=2*x*y+(x*y)^3$

$\text{diff}(f,y,1)$

$\text{diff}(f,y,2)$

$\text{diff}(f,y,4)$

L05_Q22: Find the integral of the function $f(x) = x^2 + 10x + 4$ using int command. Plot the multiple graphs of the function and its output using plot command. Label the horizontal axis with x and vertical axis with y. Keeps the title "Graph of function and its integral"

%Find the integral of the function $f(x)=x^2+10x+4$ using int command.

%Plot the multiple graphs of the function and its output using plot command.

%Label the horizontal axis with x and vertical axis with y.

%Keeps the title "Graph of function and its integral"

clc

clear all

close all

syms x

$f=\text{int}(x^2+10*x+4)$

$x=-2:0.01:2;$

$\text{plot}(x,x.^2+10.*x+4,(x.*(x.^2 + 15.*x + 12))./3,x)$

$\text{xlabel}('x\text{-axis}')$

$\text{ylabel}('y\text{-axis}')$

$\text{title}('Graph of function and its integral')$

grid

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L05_Q23: Differentiate the following functions using diff command. Moreover plot the multiple graphs of the given functions and their derivatives on the interval [0, 5]. $f(x) = x^2 + 2x + 10$ and $g(x) = \sin(x)$ **(P#116 – LECT#05)**

%Differentiate the following functions using diff command.

%Moreover plot the multiple graphs of the given functions and their derivatives on the interval [0, 5].

% $f(x)=x^2+2x+10$ and $g(x)=\sin(x)$

clc

clear all

close all

syms x

$f=x^2+2*x+10$

$g=\sin(x)$

$f1=\text{diff}(f)$

$g1=\text{diff}(g)$

$x=0:0.1:5;$

$\text{plot}(x,(x.^2+2.*x+10),x,(2.*x + 2),x,\sin(x),x,\cos(x))$

$\text{xlabel}('x\text{-axis}')$

$\text{ylabel}('y\text{-axis}')$

$\text{title}('Graph of function and its derivatives using diff and plot')$

grid

L06_Q24: Compute the values of $\sin(x)$ at $x = 0; 1$, and 2 using for loop.

(P#170 – LECT#06)

%Compute the values of $\sin(x)$ at $x = 0; 1$, and 2 using for loop.

clear all

close all

clc

for $i=1:5$

$y=\sin(i)$

end

L06_Q25: Solve the following algebraic equations using Script M-Files.

$$x^2 + 2x - 4 = 0 \text{ and } x^3 + 5x^2 + 4x + 3 = 0$$

Solve the following equations for x using Script M-Files.

$$x + \log(y) = 3 \text{ and } x + 10y = 6$$

Solve the following system of equations using Script M-Files.

$$x + y^2 = 2, y - 3x = 7 \text{ and } x + y^2 = 2, 2y^2 - 3x = 7$$

(P#173 - EXERCISE – LECT#06)

%Solve the following algebraic equations using Script M-Files.

% $x^2+2x-4=0$ and $x^3+5x^2+4x+3=0$

%Solve the following equations for x using Script M-Files. $x+\log(y)=3$ and $x+10y=6$

%Solve the following system of equations using Script M-Files.

% $x+y^2=2,y-3x=7$ and $x+y^2=2,2y^2-3x=7$

clear all

close all

clc

syms x y

$[x]=\text{solve}('x^2+2*x-4=0','x^3+5*x^2+4*x+3=0')$

$[x,y]=\text{solve}('x+\log(y)=3','x+10*y=6');$

$[x,y]=\text{solve}('x+y^2=2','y-3*x=7');$

$[x,y]=\text{solve}('x+y^2=2','2*y^2-3*x=7');$

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MIX_Q26: Find the first and second derivative of the following function with respect to x using MATLAB command. Then create a surface plot of the given function and its first and second derivatives by defining X and Y using mesh grid command with hold on and hold off command. $f(x, y) = xy(x^2 - y^2)$. (ASSIGNMENT#01)

% ASSIGNMENT#1 BY ME

% FIND FIRST & SECOND DERIVATIVES OF $f(x,y)=xy(x^2-y^2)$ w.r.t x.

% Then create surface plot of given function, its first & second derivatives using

% meshgrid command with hold on and hold off command.

clear all;

clc;

close all;

syms x y;

f=x*y*(x^2-y^2)

df=diff(f)

ddf=diff(df)

u=-2:0.1:2;

v=-2:0.1:2;

[x,y]=meshgrid(u,v);

f=eval(vectorize(f));

df=eval(vectorize(df));

ddf=eval(vectorize(ddf));

surf(x,y,f)

hold on

surf(x,y,df)

hold on

surf(x,y,ddf)

hold off

MIX_Q27: Solve: $f(x, y) = xy^2 + 2x + 4 = 0$, $f(x, y) = 3x^2y + 2xy^2 + 4x$ at $x=5$,
 $f(x) = \sin x$ at $x = 5$

%Solve: $f(x,y)=xy^2+2x+4=0$, $f(x,y)=3x^2y+2xy^2+4x$ at $x=5$

% $f(x)=\sin x$ at $x=5$

clc

clear all

close all

syms x y u v

[x,y]=solve('x*y^2+2*x+4=0')

u=3*x^2*y+2*x*y^2+4*x;

subs(u,x,5)

v=sin(x);

subs(v,x,5)

MIX_Q28: Plot the contour plots of the $\sin(x) + \cos(y)$ where x range from -2π to 2π , y range from 0 to 4π . Label the horizontal axis with x and vertical axis with y. Keep the title of the graph "Contour Plot of $\sin(X)+\cos(Y)$ ".

%Plot the contour plots of the $\sin(x) + \cos(y)$

%where x range from -2? to 2?, y range from 0 to 4?.

%Label the horizontal axis with x and vertical axis with y.

%Keep the title of the graph "Contour Plot of $\sin(X)+\cos(Y)$ ".

clc

clear all

close all

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```
syms x y
[X,Y]=meshgrid(-2*pi:0.01:2*pi,0:0.01:4*pi);
contour(X,Y,sin(X)+cos(Y))
xlabel('x-axis');
ylabel('y-axis');
title('Contour Plot of Sin(X)+Cos(Y)')
grid
```

L07_Q29: Find the solution of the following initial value problem $dy/dx = x^2 + y$; $y(1) = 3$. Moreover, plot the graph of the solution using ezplot command on the interval $[0,2]$.

(P#187-189 - EXERCISE – LECT#07)

%Find the solution of the following initial value problem $dy/dx = x^2 + y$; $y(1) = 3$.
%Moreover, plot the graph of the solution using ezplot command on the interval $[0,2]$.

```
clc
clear all
close all
syms y
eq1='Dy=x^2+y';
y=dsolve(eq1,'y(1)=3','x')
ezplot(y,[0,2])
xlabel('x-axis')
ylabel('y-axis')
title('Solution of Initial Value Problem and ezplot of solution')
grid
```

L07_Q30: Find general solution of the following boundary value problem $y'(x) + 8y'(x) + 2y(x) = \cos(x)$. Then plot the solution curve on the interval $[0,4]$ corresponding to the following boundary conditions. $y(0) = 4, y'(1) = 5$.

(P#215 - EXERCISE – LECT#07)

%Find general solution of the following boundary value problem $y''(x)+8y'(x)+2y(x)=\cos(x)$.
%Then plot the solution curve on the interval $[0,4]$
%corresponding to the following boundary conditions. $y(0)=4,y'(1)=5$.

```
clc
clear all
close all
%syms x y
eq1='D2y+8*Dy+2*y=cos(x)';
icon='y(0)=4,Dy(1)=5';
y=dsolve(eq1,icon,'x')
ezplot(y,[0,4])
xlabel('x-axis')
ylabel('y-axis')
title('Solution of Boundary Value Problem and Solution Curve')
grid
```

L07_Q31: Plot the family of curves of the following differential equation on the interval $[0,3]$, with initial values $y(0) = -3 : 3$, $dy/dx = x * y + 2 * x^2$.

(P#194-196 - EXERCISE – LECT#07)

%Plot the family of curves of the following differential equation on the interval $[0,3]$.
%with initial values $y(0) = -3 : 3$, $dy/dx = x * y + 2 * x^2$.

```
clc
clear all
```



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```
close all
eq1='Dy=x*y+2*x^2';
y=dsolve(eq1,'y(0)=c','x')
figure; hold on
for cval=-3:3
    ezplot(subs(y,'c',cval),[0,3])
end
xlabel('x-axis')
ylabel('y-axis')
title('Family of Curves of DE')
grid
```

L07_Q32: Find the general solution of the following differential equation $y'(x) = 2xy + 2x^3$. Also plot the family of solution curves on the interval $[-4,4]$ of above differential equation obtained using the following initial conditions $y(0) = -3, -2, -1, 0, 1, 2, 3$.

(P#214 - EXERCISE – LECT#07)

%Find the general solution of the following differential equation $y'(x)=2xy+2x^3$.
%Also plot the family of solution curves on the interval $[-4,4]$ of above differential equation
%obtained using the following initial conditions $y(0) = -3, -2, -1, 0, 1, 2, 3$.

```
clc
clear all
close all
eq1='Dy=2*x*y+2*x^3';
y=dsolve(eq1,'y(0)=c','x')
figure; hold on
for cval=-3:3
    ezplot(subs(y,'c',cval),[-4,4])
end
xlabel('x-axis')
ylabel('y-axis')
title('Family of Curves of DE')
grid
```

L07_Q33: Solve the following second-order differential equation $y''(x) + y'(x) - 6y = 0$ subject to the following initial conditions $y(0) = 1, y'(0) = 0$. Moreover plot the solution curve on the interval $[0,2]$. **(P#198-200 - EXERCISE – LECT#07)**

%Solve the following second-order differential equation $y''(x)+y'(x)-6y=0$
%subject to the following initial conditions $y(0)=1, y'(0)=0$.
%Moreover plot the solution curve on the interval $[0,2]$

```
clc
clear all
close all
eq1='D2y+Dy-6*y=0';
incond='y(0)=1,Dy(0)=0';
y=dsolve(eq1,incond,'x')
ezplot(y,[0,2])
xlabel('x-axis')
ylabel('y-axis')
title('Solution of 2nd Order DE')
grid
```

**Important and Solved Questions for Mid Term Exam**

L07_Q34: Solve the following second-order differential equation $y''(x) + y'(x) - 6y = 0$ subject to the following initial conditions $y(0) = 1, y'(1) = 0$. Moreover plot the solution curve on the interval $[0,3]$. **(P#201-203 - EXERCISE – LECT#07)**

%Solve the following second-order differential equation $y''(x)+y'(x)-6y=0$

%subject to the following initial conditions $y(0)=1,y'(1)=0$.

%Moreover plot the solution curve on the interval $[0,3]$.

clc

clear all

close all

eq1='D2y+Dy-6*y=0';

incond='y(0)=1,Dy(1)=0';

y=dsolve(eq1,incond,'x')

ezplot(y,[0,3])

xlabel('x-axis')

ylabel('y-axis')

title('Solution of 2nd Order DE')

grid

L07_Q35: Solve the following system of differential equations by using dsolve command.

$$x'(t) = x(t) + 2y(t) - z(t)$$

$$y'(t) = x(t) + z(t)$$

$$z'(t) = 4x(t) - 4y(t) + 5z(t)$$

(P#204-205 - EXERCISE – LECT#07)

%Solve the following system of differential equations by using dsolve.

% $x'(t)=x(t)+2y(t)-z(t)$

% $y'(t)=x(t)+z(t)$

% $z'(t)=4x(t)-4y(t)+5z(t)$

eq1='Dx=x+2*y-z';

eq2='Dy=x+z';

eq3='Dz=4*x-4*y+5*z';

[x,y,z]=dsolve(eq1,eq2,eq3)

L07_Q36: Solve the following system of differential equations.

$$x'(t) = x(t) + 2y(t) - z(t)$$

$$y'(t) = x(t) + z(t)$$

$$z'(t) = 4x(t) - 4y(t) + 5z(t)$$

Corresponding to the following conditions $x(0)=1,y(0)=2,z(0)=3$. Moreover plot the solution curves using the plot command. **(P#206-209 - EXERCISE – LECT#07)**

%Solve the following system of differential equations.

% $x'(t)=x(t)+2y(t)-z(t)$

% $y'(t)=x(t)+z(t)$

% $z'(t)=4x(t)-4y(t)+5z(t)$

%Corresponding to the following conditions $x(0)=1,y(0)=2,z(0)=3$

%Moreover plot the solution curves using the plot command

%Solve the following system of differential equations.

% $x'(t)=x(t)+2y(t)-z(t)$

% $y'(t)=x(t)+z(t)$

% $z'(t)=4x(t)-4y(t)+5z(t)$

%Corresponding to the following conditions $x(0)=1,y(0)=2,z(0)=3$

%Moreover plot the solution curves using the plot command



Important and Solved Questions for Mid Term Exam

```
clc
clear all
close all
eq1='Dx=x+2*y-z';
eq2='Dy=x+z';
eq3='Dz=4*x-4*y+5*z';
incond='x(0)=1,y(0)=2,z(0)=3';
[x,y,z]=dsolve(eq1,eq2,eq3,incond)
t=linspace(0,0.5,25);
xx=eval(vectorize(x));
yy=eval(vectorize(y));
zz=eval(vectorize(z));
plot(t,xx,t,yy,t,zz)
xlabel('x-axis')
ylabel('y-axis')
title('Graph of solution curves of DE')
grid
```

L07_Q37: Solve the following system of differential equations.

$$x''(t) = x'(t) + 2y'(t) - z'(t)$$

$$y''(t) = x'(t) + z'(t)$$

$$z''(t) = 4x'(t) - 4y'(t) + 5z'(t)$$

Corresponding to the following conditions $x(0) = 1, x'(0) = 2, y(0) = 1, y'(0) = 2, z(0) = 3, z'(0) = 4$.

Moreover plot the solution curves using the plot command.

(P#210-213 - EXERCISE – LECT#07)

%Solve the following system of differential equations.

%x''(t)=x'(t)+2y'(t)-z'(t)

%y''(t)=x'(t)+z'(t)

%z''(t)=4x'(t)-4y'(t)+5z'(t)

%Corresponding to the following conditions $x(0)=1, x'(0)=2, y(0)=1, y'(0)=2, z(0)=3, z'(0)=4$.

%Moreover plot the solution curves using the plot command.

```
clc
clear all
close all
eq1='D2x=Dx+2*y-Dz';
eq2='D2y=Dx+Dz';
eq3='D2z=4*Dx-4*y+5*z';
incond='x(0)=1,Dx(0)=2,y(0)=1,Dy(0)=2,z(0)=3,Dz(0)=4';
[x,y,z]=dsolve(eq1,eq2,eq3,incond)
t=linspace(0,5,25);
xx=eval(vectorize(x));
yy=eval(vectorize(y));
zz=eval(vectorize(z));
plot(t,xx,t,yy,t,zz)
xlabel('x-axis')
ylabel('y-axis')
title('Graph of solution curves of DE')
grid
```

**Important and Solved Questions for Mid Term Exam**

L07_Q38: Solve the following system of differential equations.

$$x''(t) = 2x'(t) + 2y'(t) - z'(t)$$

$$y''(t) = 3x'(t) - 3z'(t)$$

$$z''(t) = 5x'(t) - 4y'(t) + 5z'(t)$$

Corresponding to the following conditions $x(0) = 2, x'(0) = 1, y(0) = 3, y'(0) = 4, z(0) = 3, z'(0) = 5$ **(P#216 - EXERCISE – LECT#07)**

%Solve the following system of differential equations.

$$\%x''(t)=2x'(t)+2y'(t)-z'(t)$$

$$\%y''(t)=3x'(t)-3z'(t)$$

$$\%z''(t)=5x'(t)-4y'(t)+5z'(t)$$

%Corresponding to the following conditions $x(0)=2, x'(0)=1, y(0)=3, y'(0)=4, z(0)=3, z'(0)=5$

clc

clear all

close all

$$\text{eq1}='D2x=2*Dx+2*Dy-Dz'$$

$$\text{eq2}='D2y=3*Dx-3*Dz';$$

$$\text{eq3}='D2z=5*Dx-4*Dy+5*Dz';$$

$$\text{incond}='x(0)=2,Dx(0)=1,y(0)=3,Dy(0)=4,z(0)=3,Dz(0)=5';$$

$$[x,y,z]=\text{dsolve}(\text{eq1},\text{eq2},\text{eq3},\text{incond})$$

$$t=\text{linspace}(0,5,25);$$

$$xx=\text{eval}(\text{vectorize}(x));$$

$$yy=\text{eval}(\text{vectorize}(y));$$

$$zz=\text{eval}(\text{vectorize}(z));$$

$$\text{plot}(t,xx,t,yy,t,zz)$$

$$\text{xlabel}('x\text{-axis}')$$

$$\text{ylabel}('y\text{-axis}')$$

$$\text{title}('Graph of solution curves of DE')$$

grid

L07_Q39: Solve the following system of differential equations.

$$x'(t) = 2x(t) + 2y(t) - 3z(t)$$

$$y'(t) = 2x(t) + 3z(t)$$

$$z'(t) = x(t) - y(t) + z(t)$$

Corresponding to the following conditions $x(0)=2, y(0)=1, z(0)=4$. Moreover plot the solution curves using plot command. **(P#217 - EXERCISE – LECT#07)**

%Solve the following system of differential equations.

$$\%x'(t)=2x(t)+2y(t)-3z(t)$$

$$\%y'(t)=2x(t)+3z(t)$$

$$\%z'(t)=x(t)-y(t)+z(t)$$

%Corresponding to the following conditions $x(0)=2, y(0)=1, z(0)=4$.

%Moreover plot the solution curves using plot command.

clc

clear all

close all

$$\text{eq1}='Dx=2*x+2*y-3*z'$$

$$\text{eq2}='Dy=2*x+3*z';$$

$$\text{eq3}='Dz=x-y+z';$$

$$\text{incond}='x(0)=2,y(0)=1,z(0)=4';$$

$$[x,y,z]=\text{dsolve}(\text{eq1},\text{eq2},\text{eq3},\text{incond})$$

$$t=\text{linspace}(0,5,25);$$

$$xx=\text{eval}(\text{vectorize}(x));$$



Important and Solved Questions for Mid Term Exam

```
yy=eval(vectorize(y));  
zz=eval(vectorize(z));  
plot(t,xx,t,yy,t,zz)  
xlabel('x-axis')  
ylabel('y-axis')  
title('Graph of solution curves of DE')  
grid
```

L08_Q40: Find Numerical solution of the following differential equation on the interval [0,1] using Euler's Formula for $n=10$, $\frac{dy}{dx} = \sin(xy)$, $y(0) = \pi$ (**P#225-229 – LECT#08**)

```
function [xvalues,yvalues]=euler(f,x0,y0,xn,n);  
format long  
x0=0;  
y0=pi;  
xn=1;  
n=10;  
x(1)=x0;  
y(1)=y0;  
xn=1;  
dx=(xn-x0)/n;  
for k=1:n  
    x(k+1)=x(k)+dx;  
    y(k+1)=y(k)+sin(x(k)*y(k))*dx;  
end  
xvalues=x'  
yvalues=y'  
plot(xvalues,yvalues)
```

L08_Q41: Find the numerical solution of the following differential equation on the interval [0; 1] using Euler's method for $n = 20$. $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 0.5$

(**FOLLOW P#225-229 – LECT#08**)

%Find the numerical solution of the following differential equation on the interval [0,1]

%using Euler's method for $n = 20$. $dy/dx=(y-x)/(y+x)$, $y(0)=0.5$

```
function [xvalues,yvalues]=euler(f,x0,y0,xn,n);  
format long  
x0=0;  
y0=0.5;  
xn=1;  
n=20;  
x(1)=x0;  
y(1)=y0;  
xn=1;  
dx=(xn-x0)/n;  
for k=1:n  
    x(k+1)=x(k)+dx;  
    y(k+1)=y(k)+((y(k)-x(k))/(y(k)+x(k)))*dx;  
end  
xvalues=x'  
yvalues=y'  
plot(xvalues,yvalues)
```

**Important and Solved Questions for Mid Term Exam**

L08_Q42: Find the numerical solution of the following differential equation on the interval $[0, 1]$ using improved Euler's Formula for $n=10$, $\frac{dy}{dx} = \sin(xy)$, $y(0) = \pi$

(P#231-235 – LECT#08)

%Find the numerical solution of the following differential equation on the interval $[0,1]$

%using improved Euler's Formula for $n=10$, $dy/dx=\sin(xy)$, $y(0)=\pi$

function [xvalues,yvalues]=impeuler(f,x0,y0,xn,n);

format long

f=@(x,y)sin(x.*y);

x0=0;

y0=pi;

xn=1;

n=10;

x(1)=x0;

y(1)=y0;

xn=1;

dx=(xn-x0)/n;

for k=1:n

x(k+1)=x(k)+dx;

y(k+1)=y(k)+dx/2*(f(x(k),y(k))+f(x(k),y(k)+dx*y(k)));

end

xvalues=x'

yvalues=y'

plot(xvalues,yvalues)

L08_Q43: Find the numerical solution of the following differential equation on the interval $[0,1]$ using Improved Euler's method for $n = 20$. $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 0.5$

(FOLLOW P#231-235 – LECT#08)

%Find the numerical solution of the following differential equation on the interval $[0,1]$

%using Improved Euler's method for $n = 20$. $dy/dx=(y-x)/(y+x)$, $y(0)=0.5$

function [xvalues,yvalues]=impeuler(f,x0,y0,xn,n);

format long

f=@(x,y)((y-x)/(y+x));

x0=0;

y0=1;

xn=0.5;

n=20;

x(1)=x0;

y(1)=y0;

xn=1;

dx=(xn-x0)/n;

for k=1:n

x(k+1)=x(k)+dx;

y(k+1)=y(k)+dx/2*(f(x(k),y(k))+f(x(k),y(k)+dx*y(k)));

end

xvalues=x'

yvalues=y'

plot(xvalues,yvalues)

**Important and Solved Questions for Mid Term Exam**

L08_Q44: Find the numerical solution of the following initial value problem on the interval $[0,1]$ using improved Euler's method by taking step size $\Delta t = 0.01$, Moreover, plot the graph of the approximated solution curve. $\frac{dy}{dt} = \frac{y-t}{y+t}$, $y(0) = 0.5$

Note: The iterative form of the improved Euler's method is given below;

$$y_{k+1} \approx y_k + \frac{\Delta t}{2} (y'_k + f(t_{k+1}, y_k + \Delta t y'_k)), k = 0, 1, 2, \dots, n-1$$

(FOLLOW P#231-235 – LECT#08) –

(ABU SULTAN MID TERM Q#1 – 26 DEC 2019 – 12:00 pm)

%Find the numerical solution of the following initial value problem on the interval $[0,1]$

%using improved Euler's method by taking step size $\Delta t=0.01$,

%Moreover, plot the graph of the approximated solution curve. $dy/dt=(y-t)/(y+t)$, $y(0)=0.5$

function [yvalues,tvalues]=impeuler(f,y0,t0,yn,n);

format long

f=@(y,t)((y-t)/(y+t));

y0=0;

t0=1;

yn=0.5;

n=20;

y(1)=y0;

t(1)=t0;

yn=1;

dt=0.01;

for k=1:n

y(k+1)=y(k)+dt;

t(k+1)=t(k)+dt/2*(f(y(k),t(k))+f(y(k),t(k)+dt*t(k)));

end

yvalues=y'

tvalues=t'

plot(yvalues,tvalues)

L08_Q45: Find the numerical solution of the following differential equation on the interval $[0, 1]$ using Runge–Kutta method of fourth order for $n=10$, $\frac{dy}{dx} = \sin(xy)$, $y(0) = \pi$

(SOLUTION P#237-244 - LECT#08)

%Find the numerical solution of the following differential equation on the interval $[0,1]$

%using Runge–Kutta method of fourth order for $n=10$, $dy/dx=\sin(xy)$, $y(0)=\pi$

function [xvalues,yvalues]=RungeKutta(f,x0,y0,xn,n);

format long

f=@(x,y)sin(x.*y);

x0=0;

y0=pi;

xn=1;

n=10;

x(1)=x0;

y(1)=y0;

dx=(xn-x0)/n;

for k=1:n

x(k+1)=x(k)+dx;

k1=f(x(k),y(k));

k2=f(x(k)+dx/2,y(k)+dx/2*(k1));

k3=f(x(k)+dx/2,y(k)+dx/2*(k2));

k4=f(x(k+1),y(k)+dx*(k3));



Important and Solved Questions for Mid Term Exam

```
y(k+1)=y(k)+dx/6*(k1+2*k2+2*k3+k4);  
end  
xvalues=x'  
yvalues=y'  
plot(xvalues,yvalues)  
title('Runge Kutta Method of order four');  
grid;  
xlabel('x-axis');  
ylabel('y-axis');
```

L08_Q46: $\frac{dy}{dx} = \cos(x + y)$, $n = 10$, $[0,1]$, $y(0) = \pi$. Solve by using Range-kutta method.

(FOLLOW SOLUTION P#237-244 - LECT#08)

```
% dy/dx=Cos(x+y),n=10,[0,1],y(0)=pi  
% Solve by using Range-kutta method.  
function [xvalues,yvalues]=RungeKutta(f,x0,y0,xn,n);  
format long  
f=@(x,y)cos(x+y);  
x0=0;  
y0=pi;  
xn=1;  
n=10;  
x(1)=x0;  
y(1)=y0;  
dx=(xn-x0)/n;  
for k=1:n  
    x(k+1)=x(k)+dx;  
    k1=f(x(k),y(k));  
    k2=f(x(k)+dx/2,y(k)+dx/2*(k1));  
    k3=f(x(k)+dx/2,y(k)+dx/2*(k2));  
    k4=f(x(k+1),y(k)+dx*(k3));  
    y(k+1)=y(k)+dx/6*(k1+2*k2+2*k3+k4);  
end  
xvalues=x'  
yvalues=y'  
plot(xvalues,yvalues)  
title('Runge Kutta Method of order four');  
grid;  
xlabel('x-axis');  
ylabel('y-axis');
```

L08_Q47: Find the numerical solution of the following differential equation on the interval $[0,1]$ using Runge–Kutta method of fourth order for $n = 20$ $\frac{dy}{dt} = \frac{y-x}{y+x}$, $y(0) = 0.5$

(FOLLOW SOLUTION P#237-244 - LECT#08)

```
% Find the numerical solution of the following differential equation on  
% the interval [0, 1] using Runge–Kutta method of fourth order for n = 20  
% dy/dt=(y-x)/(y+x) ,y(0)=0.5  
function [xvalues,yvalues]=RungeKutta(f,x0,y0,xn,n);  
format long  
f=@(x,y)((y-x)/(y+x));  
x0=0;  
y0=0.5;
```



Important and Solved Questions for Mid Term Exam

```
xn=1;
n=20;
x(1)=x0;
y(1)=y0;
dx=(xn-x0)/n;
for k=1:n
    x(k+1)=x(k)+dx;
    k1=f(x(k),y(k));
    k2=f(x(k)+dx/2,y(k)+dx/2*(k1));
    k3=f(x(k)+dx/2,y(k)+dx/2*(k2));
    k4=f(x(k+1),y(k)+dx*(k3));
    y(k+1)=y(k)+dx/6*(k1+2*k2+2*k3+k4);
end
xvalues=x'
yvalues=y'
plot(xvalues,yvalues)
title('Runge Kutta Method of order four');
grid;
xlabel('x-axis');
ylabel('y-axis');
```

L08_Q48: Find Numerical solution of the following differential equation on the interval $[0,1]$ using Taylor's Formula for $n=10$, $\frac{dy}{dx} = \sin(xy)$, $y(0) = \pi$

(SOLUTION P#245-250 - LECT#08)

%tical solution of the following differential equation on the interval

%[0,1] using Taylor's Formula for $n=10$, $dy/dx=\sin(xy)$, $y(0)=?$

```
function [xvalues,yvalues]=TaylorOrder(f,x0,y0,xn,n);
```

```
format long
```

```
f=@(x,y)sin(x.*y);
```

```
x0=0;
```

```
y0=pi;
```

```
xn=1;
```

```
n=10;
```

```
x(1)=x0;
```

```
y(1)=y0;
```

```
dx=(xn-x0)/n;
```

```
for k=1:n
```

```
    x(k+1)=x(k)+dx;
```

```
    y(k+1)=y(k)+f(x(k),y(k))*dx+(y(k)*cos(x(k)*y(k))+x(k)*cos(x(k)*y(k))*f(x(k),y(k)))*(dx)^2/2;
```

```
end
```

```
xvalues=x'
```

```
yvalues=y'
```

```
plot(xvalues,yvalues)
```

```
title('Taylor's Method of Second Order');
```

```
grid;
```

```
xlabel('x-axis');
```

```
ylabel('y-axis');
```

**Important and Solved Questions for Mid Term Exam**

L08_Q49: Find the numerical solution of the following differential equation on the interval $[0,1]$ using Taylor's method of second order for $n = 20$. $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 0.5$

(FOLLOW SOLUTION P#245-250 - LECT#08)

%Find the numerical solution of the following differential equation on

% the interval $[0,1]$ using Taylor's method of second order for $n = 20$.

% $dy/dx=(y-x)/(y+x)$, $y(0)=0.5$

function [xvalues,yvalues]=TaylorOrder(f,x0,y0,xn,n);

format long

$f=@(x,y)((y-x)/(y+x));$

$x0=0;$

$y0=0.5;$

$xn=1;$

$n=20;$

$x(1)=x0;$

$y(1)=y0;$

$dx=(xn-x0)/n;$

for $k=1:n$

$x(k+1)=x(k)+dx;$

$y(k+1)=y(k)+f(x(k),y(k))*dx+(y(k)*\cos(x(k)*y(k))+x(k)*\cos(x(k)*y(k))*f(x(k),y(k)))*(dx)^2/2;$

end

$xvalues=x'$

$yvalues=y'$

$plot(xvalues,yvalues)$

$title('Taylor's Method of Second Order');$

$grid;$

$xlabel('x-axis');$

$ylabel('y-axis');$

L08_Q50: Solve by Taylor method $\frac{dy}{dx} = \cos(xy)$, $y(0) = 1$, here $n = 10$ order.

(FOLLOW SOLUTION P#245-250 - LECT#08)

%Solve by Taylor method $dy/dx=\cos(xy)$, $y(0)=1$, here $n=10$ order

function [xvalues,yvalues]=TaylorOrder(f,x0,y0,xn,n);

format long

$f=@(x,y)\cos(x.*y);$

$x0=0;$

$y0=1;$

$xn=1;$

$n=10;$

$x(1)=x0;$

$y(1)=y0;$

$dx=(xn-x0)/n;$

for $k=1:n$

$x(k+1)=x(k)+dx;$

$y(k+1)=y(k)+f(x(k),y(k))*dx+(y(k)*\cos(x(k)*y(k))+x(k)*\cos(x(k)*y(k))*f(x(k),y(k)))*(dx)^2/2;$

end

$xvalues=x'$

$yvalues=y'$

$plot(xvalues,yvalues)$

$title('Taylor's Method of Second Order');$



Important and Solved Questions for Mid Term Exam

```
grid;  
xlabel('x-axis');  
ylabel('y-axis');
```

MIX_Q51: Find the solution of the following second order differential equation

$$y''(t) + 8y'(t) + 2y(x) = \cos(x). \quad (\text{PAST PAPERS})$$

%Find the solution of the following second order differential equation

```
%y''(t)+8y'(t)+2y(x)=Cos(x) .
```

```
clc
```

```
clear all
```

```
close all
```

```
eq1='D2y+8*Dy+2*y(x)=cos(x) '
```

```
dsolve(eq1, 't')
```

MIX_Q52: Plt to the contour plots of the expression $z=x*\exp(-x^2-y^2)$ range is $-2 < x < 2$ and $-3 < y < 3$. Label the horizontal axis with x and vertical axis with y. keep the title of graph contour plots of the expression.

(ABU SULTAN MID TERM Q#2 – 26 DEC 2019 – 12:00 pm)

```
%Plt to the contour plots of the expression z=x*exp(-x^2-y^2)
```

```
%range is -2<x<2 and -3<y<3
```

```
%Label the horizontal axis with x and vertical axis with y
```

```
%keep the title of graph contour plots of the expression
```

```
clc
```

```
clear all
```

```
close all
```

```
syms x y
```

```
[X,Y]=meshgrid(-2:0.5:2,-3:0.5:3);
```

```
contour(X,Y,X.*exp(-X.^2-Y.^2));
```

```
xlabel('x-axis');
```

```
ylabel('y-axis');
```

```
title('Graph of Contour plots of the expression');
```

```
grid;
```

The End

**Don't look for someone who can solve your problems,
Instead go and stand in front of the mirror,
Look straight into your eyes,
And you will see the best person who can solve your problems!
Always trust yourself.**

(BY ABU SULTAN)